Safety Data Sheet (SDS)

Citric acid monohydrate

According to 1907/2006/EC, Article 31

Printing date 29.03.2018

Revision: 29.03.2018

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

1.1 Product identifiers

Chemical Name	:	Citric acid
Trade name	:	Citric Acid Monohydrate
CAS-No	:	5949-29-1

1.2 Details of the supplier of the safety data sheet

Company	:	SUNSHINE BIOTECH INTERNATIONAL CO.,LTD. 188 MOO 4 TAMBOL THATOOM, SRIMAHAPHOTE,PRACHINBURI 25140 THAILAND
Telephone	:	037278801
Email	:	sunshinethaixsb@thaisunshine.net
Website	:	www.sunshinethai.net

1.3 Emergency telephone number

Emergency Phone # : 037278802 (8:00 **am. -** 5:00 **pm.) [Office hours]**

SECTION 2: HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008

Eye irritation (Category 2), H319

For the full text of the H-Statements mentioned in this Section, see Section 16.

CLASSIFICATION according to EU directives 67/548/EEC or 1999/45/EC

XI IRRITANT R36

For the full text of the R-phrases mentioned in this Section, see Section 16.

2.2 Label elements

Labelling according Regulation (EC) No 1272/2008

Pictogram



GHS07

Signal word	Warning
Hazard statement(s)	
H319	Causes serious eye irritation.
Precautionary statement(s)	
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
Supplemental Hazard Statements	none

2.3 Other hazards – none

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Chemical characterization: Substances

Formula	:	$C_6H_8O_7 \cdot H_2O$
Molecular Weight	:	210,14 g/mol
CAS-No.	:	5949-29-1
Identification number(s)		
EC-No.	:	201-069-1

SECTION 4: FIRST AID MEASURES

4.1 Description of first aid measures

General information:

Immediately remove any clothing soiled by the product.

Personal protection for the First Aider.

Take affected persons out into the fresh air.

After inhalation: Supply fresh air; consult doctor in case of complaints.

After skin contact:

Wash off with plenty of water.

Immediately wash with water and soap and rinse thoroughly.

Immediately remove any clothing soiled by the product.

If skin irritation continues, consult a doctor.

After eye contact:

Rinse opened eye for several minutes under running water.

Call a doctor immediately.

After swallowing:

make victim drink water (maximum of 2 drinking glasses)

If symptoms persist consult doctor.

4.2 Most important symptoms and effects, both acute and delayed

No further relevant information available.

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available.

SECTION 5: FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable extinguishing agents:

CO_2, powder or water spray. Fight larger fires with water spray or alcohol resistant foam.

For safety reasons unsuitable extinguishing agents: Water with full jet

5.2 Special hazards arising from the substance or mixture

Combustible.

Formation of toxic gases is possible during heating or in case of fire.

In case of fire, the following can be released:

carbon oxides (CO, CO_2).

5.3 Advice for firefighters

Protective equipment: Wear self-contained respiratory protective device.

Additional information

Dispose of fire debris and contaminated fire fighting water in accordance with official regulations.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Avoid formation of dust.

Do not inhale dust.

Avoid substance contact.

Ensure adequate ventilation

6.2 Environmental precautions: Do not allow to enter sewers/ surface or ground water.

6.3 Methods and material for containment and cleaning up:

Pick up mechanically.

Avoid generation of dusts.

Clean up affected area.

6.4 Reference to other sections

See Section 7 for information on safe handling.

See Section 8 for information on personal protection equipment.

See Section 13 for disposal information.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Any unavoidable deposit of dust must be regularly removed.

Keep receptacles tightly sealed.

Information about fire - and explosion protection: No special measures required.

7.2 Conditions for safe storage, including any incompatibilities

Storage:

Requirements to be met by storerooms and receptacles: Provide acid-resistant floor.

Information about storage in one common storage facility: Not required.

Further information about storage conditions:

Keep container tightly sealed.

Open receptacle only under localized extractor facilities.

Recommended storage temperature: +15 - +25 °C

Storage class: 11

7.3 Specific end use(s) No further relevant information available.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Additional information about design of technical facilities: No further data; see item 7.

8.1 Control parameters

Ingredients with limit values that require monitoring at the workplace: Not required

PNECs

Aquatic compartment - freshwater	0.44 mg/L
Aquatic compartment - marine water	0.044 mg/L
Aquatic compartment - sediment in freshwater	3.46 mg/kg
Aquatic compartment - sediment in marine water	34.6 mg/kg
Terrestrial compartment - soil	33.1 mg/kg
Sewage treatment plant	>1000 mg/L

Additional information: The lists valid during the making were used as basis.

8.2 Exposure controls

Personal protective equipment:

General protective and hygienic measures:

Keep away from foodstuffs, beverages and feed.

Immediately remove all soiled and contaminated clothing

Wash hands before breaks and at the end of work.

Avoid contact with the eyes.

Avoid contact with the eyes and skin.

Respiratory protection: Filter P2

Protection of hands:

The glove material has to be impermeable and resistant to the product/ the substance/ the preparation.

Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation

Material of gloves

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer.

Penetration time of glove material

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

For the permanent contact gloves made of the following materials are suitable:

Nitrile rubber, NBR

Recommended thickness of the material: $\geqslant$ 0.11 mm

Value for the permeation: Level $\geqslant$ 480 min

Eye protection:



Tightly sealed goggles

Body protection:

Protective clothing should be selected specifically for the working place, depending on concentration and quantity of the hazourdous substances handled.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

General Information

Appearance:

Form:	Solid
Colour:	White
Odour:	Odourless
Odour threshold:	Not determined.

pH-value:	1.8
Change in condition	
Melting point/freezing point:	135-152 °C
Initial boiling point and boiling range:	Undetermined.
Flash point:	Not applicable.
Flammability (solid, gas):	Product is not flammable.
Ignition temperature:	
Decomposition temperature:	Not determined.
Auto-ignition temperature:	Not determined.
Explosive properties:	Product does not present an explosion hazard.
Explosion limits:	
Lower:	Not determined.
Upper:	Not determined.
Vapour pressure:	Not applicable.
Density at 20 °C:	1.54 g/cm³
Relative density	Not determined.
Vapour density	Not applicable.
Evaporation rate	Not applicable.
Solubility in / Miscibility with water at 20 °C:	676 g/l
Partition coefficient: n-octanol/water:	Not determined.
Viscosity:	
Dynamic:	Not applicable.
Kinematic:	Not applicable.
9.2 Other information	No further relevant information available.

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity No dangerous reactions known.

10.2 Chemical stability

Thermal decomposition / conditions to be avoided: Strong heating

10.3 Possibility of hazardous reactions

Exothermic reactions with:

oxidizing agent

reducing agents

metals

Bases

10.4 Conditions to avoid Thermal decomposition: >170 °C

10.5 Incompatible materials:

oxidizing agent

Bases

reducing agents

10.6 Hazardous decomposition products: In the event of fire: See chapter 5

Additional information: releases water of crystallization when heated.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute toxicity Based on available data, the classification criteria are not met.

LD/LC50 values relevant for classification:

Quantitative data on the toxicological effect of this product are not available

Components	Type	Value	Species
Oral	LD50	5400 mg/kg	(mouse)
Dermal	LD50	>2000 mg/kg	(rat)

Primary irritant effect:

Skin corrosion/irritation Based on available data, the classification criteria are not met.

Serious eye damage/irritation

Causes serious eye irritation.

After inhalation: No irritant effect.

Respiratory or skin sensitisation Based on available data, the classification criteria are not met.

CMR effects (carcinogenity, mutagenicity and toxicity for reproduction)

Germ cell mutagenicity Based on available data, the classification criteria are not met

Carcinogenicity Based on available data, the classification criteria are not met.

Reproductive toxicity Based on available data, the classification criteria are not met.

STOT-single exposure Based on available data, the classification criteria are not met.

STOT-repeated exposure Based on available data, the classification criteria are not met.

Aspiration hazard Based on available data, the classification criteria are not met.

SECTION 12: ECOLOGICAL INFORMATION

12.1 Toxicity

Aquatic toxicity: No further relevant information available.

Type of test	Effective concentration	Method	Assessment
LC50/24 h	1535 mg/l (daphnia magna)		
LC50/48 h	440 mg/l (fish)		

12.2 Persistence and degradability The product is easily biodegradable.

12.3 Bioaccumulative potential No further relevant information available.

12.4 Mobility in soil No further relevant information available.

Additional ecological information:

General notes:

Rinse off of bigger amounts into drains or the aquatic environment may lead to decreased pH-values. A low pH-value harms aquatic organisms. In the dilution of the use-level the pH-value is considerably increased, so that after the use of the product the aqueous waste, emptied into drains, is only low water-dangerous.

Water hazard class 1 (German Regulation) (Self-assessment): slightly hazardous for water

Do not allow to enter waters, waste water, or soil.

12.5 Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

12.6 Other adverse effects No further relevant information available.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Recommendation

Chemicals must be disposed of in compliance with the respective national regulations.

Must not be disposed together with household garbage. Do not allow product to reach sewage system.

Uncleaned packaging:

Recommendation:

Disposal must be made according to official regulations.

Packagings that may not be cleansed are to be disposed of in the same manner as the product.

Recommended cleansing agents: Water, if necessary together with cleansing agents

SECTION 14: TRANSPORT INFORMATION

14.1 UN number

ADR/RID: - IMDG: - IATA: -

14.2 UN proper shipping name

ADR/RID: Not dangerous goods

IMDG: Not dangerous goods

IATA: Not dangerous goods

14.3 Transport hazard class(es)

ADR/RID: - IMDG: - IATA: -

14.4 Packaging group

ADR/RID: - IMDG: - IATA: -

14.5 Environmental hazards

ADR/RID: no IMDG Marine pollutant: no IATA: no

14.6 Special precautions for user

no data available

SECTION 15: REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Directive 2012/18/EU

Named dangerous substances - ANNEX I Substance is not listed.

15.2 Chemical safety assessment: A Chemical Safety Assessment has been carried out.

SECTION 16: OTHER INFORMATION

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship

Full text of H-Statements referred to under sections 2 and 3.

Eye Irrit.	Eye irritation
H319	Causes serious eye irritation.

Full text of R-phrases referred to under sections 2 and 3

Xi	Irritant
R36	Irritating to eyes.

Further information

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. The information in this document is based on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. It does not represent any guarantee of the properties of the product. Sunshine Biotech International Co.,LTD and its Affiliates shall not be held liable for any damage resulting from handling or from contact with the above product. See www.sunshinethai.net for additional terms and conditions of sale.